Ali Erinç Ayaz

erincayaz852@gmail.com github.com/erincayaz

Work Experience

Full-Stack Developer **Jun 2025 – Present**

Siemens — DTA OnSite

- Develop full-stack features for an industrial IoT edge platform, utilizing **Go** microservices and **React/TypeScript** to deliver real-time condition monitoring data from drivetrain equipment.
- Design and implement **gRPC** services with **Protobuf** contracts to ensure high-performance, low-latency inter-service communication within a distributed architecture.
- Build robust REST APIs using **Gin** and integrate **InfluxDB** for efficient ingestion and querying of high-frequency time-series sensor data.
- Engineer responsive frontend components using **Redux Toolkit**, **AG Grid**, and **Apache ECharts** to power real-time data visualization dashboards.
- Author comprehensive unit tests in Go with **testify** and mock repositories, maintaining strict **SonarQube** coverage gates within **GitLab CI/CD** pipelines.
- Integrate **MQTT** and **OPC UA** protocols for industrial device communication, deploying services via containerized **Docker** environments.

Full-Stack Developer **Aug 2023 – Jun 2025**

Siemens — Building X

- I joined the team at the start of the project, helping establish the backend and frontend foundations within **microservice architecture**.
- Designed and implemented features across the stack using **Node.js**, **PostgreSQL**, **Angular**, and **TypeScript** primarily for occupancy analytics and environmental quality tracking applications of Building X.
- Developed robust unit tests (**Jest**, **Angular TestBed**) and end-to-end tests using **Playwright**, fully integrated into **Gitlab CI/CD pipelines**.
- Deployed and monitored services via **Docker** containers on **AWS ECS**, using **CloudWatch** for logging and observability.
- Operated within an **agile team**, regularly presenting new features and progress to stakeholders.

Software Engineer (Part-Time) **Feb 2023 – Aug 2023**

Siemens — Industrial Edge

- Worked as a part of the Industrial Edge platform team, contributing to a **microservice-based** web application that allows factory customers to manage their deployed software on Siemens devices.
- Primarily developed backend services using **Java Spring** within a distributed architecture; also collaborated with services written in **Node.js** and **Go**.
- Maintained high test coverage with unit and feature tests (Cucumber), in line with the team's high test coverage standard.
- Developed and maintained frontend components using Angular.

Software Developer **May 2022 - Feb 2023**

23 Studios

- Developed a browser-based game for Redbull using **PlayCanvas** and **TypeScript**, with a **Node.js** backend; the game was publicly accessible on Redbull's official **website**.
- Worked as Unity developer for HitReads, an interactive book **app** built with **Unity**, which reached over 50,000 users.
- Worked in an **Agile** team environment, participating in sprint planning, regular stand-ups, and iterative development cycles to deliver features collaboratively.

Game Developer **Oct 2021 – Jan 2022**
Monster Gaming Lab

- Developed two mobile hyper-casual games, Inky Adventures and Hoppy Balls, using **Unity** and **C#**.
- Implemented core mechanics and game systems following object-oriented principles to ensure modularity and performance.
- Collaborated with a small team in a rapid prototyping cycle; both games were tested in the market by Rollic.
- **Hoppy Balls** and **Inky Adventures** are both released on App Store and downloaded by 100+ people.

Data Science Intern **Jun 2021 – Sep 2021**
Tubitak Bilgem

- Developed **CNN** and **CLDNN** models using **Keras** to classify 11 types of radio modulation.
- Surpassed accuracy of earlier internal models by integrating recent research insights.

Education

Sabanci University | *Istanbul, Tuzla* **Sep 2018 – Jun 2023**

- B.S. in Computer Science and Engineering
- Coursework: Operating Systems, Databases, Algorithms & Data Structures, Programming Languages, Computer Architecture, Computer Graphics, Computer Networks, Cyber Security

Projects

- **Automatic Anomaly Detection -** *Python*, *Scikit-Learn*, *Tensorflow*
  - Built a hybrid system using One-Class SVM and LSTM Autoencoder to detect anomalies in telemetry data provided by Koc Finansman.
- **Computer Architecture Final Project -** *Assembly*
  - Designed and implemented an encoder and decoder using MIPS assembly language. **Github Repo**
- **Maze Runner 3D Game -** *C++*, *OpenGL*
  - Developed a 3D puzzle game with custom rendering features such as lighting, shadows, shaders, and ray tracing; implemented A* pathfinding algorithm. **Github Repo**

Additional Experience

- Created an open-source Obsidian **plugin** using **TypeScript**, enabling users to customize text colors. The plugin has over 90 stars on GitHub and 250,000+ downloads, with automated deployment and testing via **GitHub Actions**.
- Developed an open-source **web automation tool** using **Playwright**, running as a scheduled **GitHub Actions** job; currently has 60+ forks, 10+ stars and a growing contributor base.
- Co-developed a multiplayer physics-based party game, **Snowbrawll**, using **Unity** and **Steam P2P Networking**.
- Completed the Inzva Algorithm Program (2020), a selective 6-month training in data structures and algorithms.
- Participated in a two-day hackathon at Peak Games (2022), implementing features for a prototype version of their mobile hit game Toon Blast (50M+ downloads).
- Released two indie games on itch.io: **Gotur** and **Kensuke**

Languages and Technologies

- **Languages**: Python, C#, C++, JavaScript, TypeScript, SQL, Go, Java
- **Frontend**: React, Angular, HTML, CSS, Redux Toolkit, AG Grid, Apache ECharts
- **Backend**: Node.js, Spring, Gin, gRPC, Protobuf, REST, MQTT, OPC UA, PostgreSQL, InfluxDB, SQLite
- **Infrastructure & Tools**: Git, Docker, AWS (ECS, S3, CloudWatch), GitLab CI/CD, GitHub Actions, SonarQube, Linux
- **Other**: Unity, Playwright, Serverless, OOP, Layered Architecture